Your Retriever Already Knows: Predicting RAG Retrieval Sufficiency from Score Distributions

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Abstract. Standard Retrieval-Augmented Generation (RAG) pipelines often provide no reliable inference-time signal of whether retrieval succeeded; on ambiguous or out-of-scope queries, generation may then hallucinate. Motivated by a Czech nuclear-regulator deployment where data sensitivity precludes third-party LLM APIs, we compare three Query Performance Prediction (QPP) paradigms for retrieval sufficiency in RAG: score-based features, a content-based LLM judge, and a hybrid. On the public ViDoRe benchmark, GeneralQPP reaches a weighted-average AUROC of 0.856, above the Classic Full QPP baseline (0.835) and the tested local multimodal LLM judge (0.649), while remaining millisecond-level. Adding the local judge as an auxiliary feature can help in selected settings, but the gain is setting-dependent and comes at LLM latency. On the private SÚJB deployment corpus, the same ranking broadly holds; the public ViDoRe benchmark and the accompanying code release provide the reproducible basis for comparison. Our conclusions are strongest for local, privacy-constrained multimodal RAG.

Keywords: Query Performance Prediction · Retrieval-Augmented Generation · Retrieval Sufficiency · Safety-Critical RAG · Vision Document Retrieval

1 Introduction

Retrieval-Augmented Generation (RAG) [11] grounds a language model on documents retrieved from a corpus. Our scope is *pre-generation retrieval-sufficiency prediction* (narrower than full answer-quality prediction): without ground-truth labels, predict whether the top-k context likely contains the answer so the system can decide at inference time whether to answer, abstain, or escalate before the LLM hallucinates. Generation-side errors, reasoning-time retrieval, and self-reflective RAG variants are complementary to this setting rather than replacements for it. In safety-critical deployments the need is acute: our motivating Czech nuclear-regulator pipeline (SÚJB) forbids third-party LLM APIs for data-sensitivity reasons, and incorrect answers carry real consequences, so a confidence estimator must be low-latency, cost-efficient, and locally deployable under confidentiality constraints. Can the *distribution shape* of retrieval scores provide a reliable enough signal without reading document content, or does content-aware judging add essential information?

2 Related Work

This work builds on Query Performance Prediction (QPP), dense-retrieval confidence estimation, RAG retrieval evaluation, and LLM-based retrieval judging.

Classic post-retrieval QPP. QPP has a long history in sparse IR, with unsupervised post-retrieval predictors derived from top-k score statistics. Three classics anchor the field: NQC [17] reads query drift from top-k standard deviation; WIG [21] contrasts top-k mean against a corpus baseline; SMV [18] combines magnitude and dispersion. These signals remain the backbone of later QPP studies, including those below.

Neural and dense QPP. Dense retrievers break some classic-predictor assumptions. Coherence-based measures [20] exploit embedding-space neighbors; projection-displacement and dimension-importance estimators [6,7,5] probe dense-representation geometry. A cross-paradigm evaluation [4] finds classic predictors fail to generalize across collections and retrievers, motivating supervised feature-rich models.

QPP for RAG. Direct application of QPP to RAG is recent. Huly et al. [9] train a supervised regressor on six post-retrieval features (WIG, NQC, SMV, top-k mean and variance, U-SMV) to predict the perplexity gain that retrieval provides for text completion. Tian et al. [19] define Retrieval and Generation Performance Prediction, combining four score-only features (NQC, MaxScore, Dense-QPP, A-Pair-Ratio) with BERT-QPP [1], a cross-encoder regressor on top-1 relevance. Both works confirm that retrieval-side features alone carry substantial signal for RAG success, but neither compares them against an LLM that actually reads the retrieved documents. Closest in spirit, Joren et al. [10] frame RAG through *sufficient context* (whether the retrieved passages suffice to answer), but operationalize it with an LLM autorater that reads the context, whereas we predict sufficiency from retrieval scores alone.

LLMs as retrieval judges. A complementary line uses LLMs to assess retrieval quality: QPP-GenRE [13] fine-tunes open-source LLMs to emit per-document relevance judgments reconstructable into classical IR metrics, and Self-RAG [2] uses reflection tokens for self-critique. Such judges can be accurate but inference cost scales with retrieval depth and context, and sensitive deployments may disallow third-party APIs entirely.

Vision document retrieval. ColPali [8] embeds document page images directly via late-interaction MaxSim, outperforming OCR pipelines. ViDoRe V3 [12] benchmarks vision RAG across eight enterprise domains. QPP for this multimodal page-image setting remains relatively unexplored, particularly for retrieval sufficiency in vision-document RAG.

3 Experimental Setup

3.1 Datasets

We evaluate on two datasets[3]: SÚJB (in-domain, Czech regulatory) and ViDoRe (cross-domain, 8 domains).

SÚJB. The production RAG corpus motivating this work: 517 pages from 5 Czech nuclear regulatory documents with 1,510 synthetic queries generated by GPT-4o from page-level OCR (stratified per document): 510 standard answerable grounded in one page, 500 paraphrases (synonym, full-rephrase, cross-lingual), and 500 adversarial unanswerable asking about nuclear-safety information absent from the corpus. We use synthetic queries because real user logs from the deployment were not available during development; we therefore treat SÚJB primarily as a controlled deployment case study rather than a substitute for live-query evaluation. A live-query benchmark for the regulator-facing deployment setting is under development. We use binary sufficient/not-sufficient labels: a page is *sufficient* iff the query is fully answerable from its content. The retriever and judge components are shown in Fig. 1. We use a fixed 80/20 stratified split (1,208 train, 302 test, seed=42) for QPP training.

ViDoRe. For ViDoRe V3 [12] we use an 8-domain benchmark setup (physics, energy, finance EN/FR, HR, industrial, computer science, pharmaceuticals) comprising 14,514 queries. V3 uses human-verified relevance labels 1 (relevant) or 2 (fully relevant, self-sufficient); unlabeled pages are treated as 0. Queries for which no fully-relevant page exists in the ViDoRe corpus count as "unanswerable" (score-1 pages may still exist); SÚJB's adversarial queries are stricter, with no corpus page relevant at all. We use stratified 80/20 splits within each domain (seed=42), training and evaluating *per* domain to measure in-domain QPP performance across diverse retrieval settings.

Reproducibility. The public GitHub code release contains the implementation, prompts, parser, feature extraction, splits, and ViDoRe evaluation scripts. Since SÚJB is private, ViDoRe is the public benchmark on which the main results can be reproduced and compared against future methods.

3.2 Ground Truth

Retrieval success is a binary Hit@k label: positive if at least one self-sufficient page appears in the top-k ($score = 1$ on SÚJB, $score \geq 2$ on ViDoRe). We report both $k = 5$ and $k = 10$ as a cutoff-robustness check; ViDoRe's eight domains test cross-distribution generalization (Section 5.4). Hit@k matches the operational RAG objective in our setting: the system needs at least one self-sufficient page to support a grounded answer, not majority recall [9].

3.3 Evaluation Metrics

Each method produces a confidence score $\hat{p}_i \in [0, 1]$ for query q_i, evaluated against binary labels $y_i \in \{0, 1\}$ with four complementary metrics. *AUROC*,

[3] GitHub repository: https://github.com/veselm73/rag-confidence.

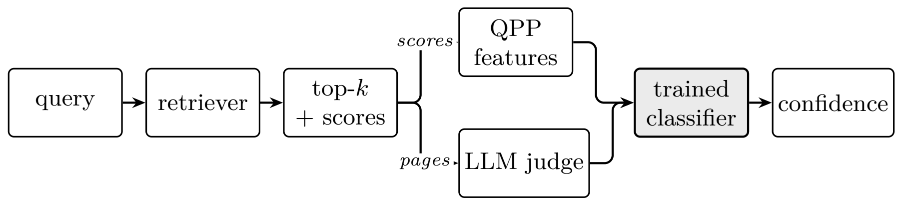

Fig. 1. Shared retrieval-sufficiency architecture: the retriever feeds score-only, content-based (LLM judge), and hybrid confidence estimators; the hybrid path combines both signals in the trained classifier.

our primary metric, measures discrimination: the probability that a random success outranks a random failure, $P(\hat{p}_i > \hat{p}_j \mid y_i = 1, y_j = 0)$. *AUPRC* is the area under the precision–recall curve, more sensitive to class imbalance. Two calibration metrics (lower is better) assess confidence quality: the Brier score [3], $\frac{1}{N}\sum_i(\hat{p}_i - y_i)^2$, and the expected calibration error ECE [14], $\sum_{b=1}^{10} \frac{|B_b|}{N} |\mathrm{freq}(B_b) - \mathrm{conf}(B_b)|$, over 10 equal-width confidence bins B_b. All results are averaged over 5 random seeds.

4 Methods

We evaluate methods from three paradigms: score-based predictors (S0-S4) that read only retrieval scores, a content-based LLM judge that reads the retrieved pages (C1), and hybrids that combine both (H1-H3); the full pipeline is shown in Fig. 1. All trained methods share the architecture-selection protocol described below (Section 4.1); per-method feature sets are then specified in Sections 4.2–4.4. The pipeline is intentionally a fixed pre-generation retrieval gate. Agentic or self-reflective RAG systems may retrieve or re-rank during reasoning; such systems are outside our experimental scope, although they could still use a retrieval-sufficiency signal as one component.

4.1 Architecture Selection

All trained methods (S1–S4, H1–H3) share a common protocol: 3-fold stratified CV on the training set selects the best architecture by AUROC, features are standardized, and the chosen model is refit on the full training fold. The candidate pool spans Logistic Regression, Ridge, and MLPs of increasing width trained with early stopping (5 candidates for smaller SÚJB dataset, 11 for ViDoRe splits, fixed a priori by training-set size; full architecture grid in the released code). Per-method CV picks are reported in Table 2. A shared architecture search controls for the fact that the baseline methods were originally paired with different classifier heads. This makes the comparison focus on feature usefulness rather than classifier capacity. In deployment, where labels may be scarce, one would fix a single model family; in our experiments this would be an MLP with little accuracy cost. C1 is not trained and has no architecture to select; it uses only a monotonic Platt-scaling wrapper [15], which preserves AUROC and AUPRC and changes only the calibration metrics (Brier, ECE). All preprocessing and model selection are performed strictly within the training fold of each split.

Table 1. GeneralQPP (S1) feature set: 24 non-lexical features in three groups.

Group	Features
Distribution (15)	top1 − p99 gap, **top1 − top10 gap**, **top-10 std. dev.**, score decay slope, **bimodal gap**, exp. decay rate, **99th percentile**, n above 0.8, n **above 0.7**, **top-50 skewness**, **top-5 concentration**, **max 2nd derivative (elbow)**, n above τ, top-1 margin over τ, gap concentration
Query (5)	**char length**, **word count**, **has numbers**, **punctuation count**, **avg. word length**
Global (4)	corpus mean sim., corpus median sim., max sim., percentile range (p90 − p10)

4.2 Score-Based Methods

S0: MaxSim Baseline. The simplest baseline uses the maximum similarity score as the confidence estimate, with no learned model.

S1: GeneralQPP (Ours; GenQPP *in tables).* We propose 24 non-lexical features organized in three groups (Table 1), trained on binary Hit@k labels with per-dataset architecture selection (Section 4.1). The 24 features target distribution *shape* (gaps, decay, skewness, bimodality, concentration), query-surface statistics, and global similarity-range baselines; only *sim_max* overlaps with the classical pool (MaxScore). The key novelty is that most features describe the geometry of the retrieval-score profile itself rather than reusing classical IR score statistics verbatim. Explicit formulas for all 24 features are provided in the code release. We additionally report S1-Lean, a 13-feature subset. Greedy forward selection and backward elimination (adding, resp. removing, one feature per step), each scored by single-seed Leave-One-Domain-Out (LODO) AUROC over the eight ViDoRe domains, both peak at the same 13 features (no fixed threshold; Section 6): 8 distribution-shape and all 5 query-surface (no global features survive). Intuitively, successful retrieval tends to produce either a peaked score profile with a clear leading page or a compact high-scoring cluster, whereas failed retrieval often produces flatter score curves. The shape-oriented features are designed to capture this geometry without inspecting document content; query-surface and global features act as lightweight controls for query form and corpus-level score scale. The query-surface features avoid vocabulary-specific lexical matching, although their distributions may still vary across languages. For the derived quantities in Table 1, τ denotes the threshold calibrated for 90% coverage; gap concentration is $(\text{top1} - \text{top2})/(\text{top1} - \text{top10})$; bimodal gap is the difference between the mean of the top three scores and the mean of ranks 10–30; and top-5 concentration is the ratio $\sum \text{top5} / \sum \text{top100}$. The bold features form the S1-Lean subset.

S2: Huly et al. SIGIR'25 [9]. Six features from the post-retrieval prediction framework: WIG, NQC, SMV, top-k mean, top-k variance, and U-SMV.

S3: Tian et al. ECIR'26 (score-only) [19]. Four score-only features from the RPP framework are NQC, MaxScore, Dense-QPP, and A-Pair-Ratio. We omit BERT-QPP (their fifth feature) as it reads document content.

S4: Classic Full. A literature baseline pooling 18 score-only QPP features into one trained predictor: Tian et al. (NQC, MaxScore, Dense-QPP, A-Pair-Ratio) [19]; Huly et al. (WIG, SMV, U-SMV, top-k mean, top-k variance) [9]; Faggioli et al. (query-centroid similarity, top-k dispersion, query drift, angular dispersion, k-NN distance, DIME-PRF concentration, PRF-DIME) [5,7,6]; Vlachou-Konchylaki et al. (top-10 coherence, score-weighted coherence) [20]. S4 controls for total feature-set size when benchmarking against GeneralQPP.

4.3 Content-Based Method

C1: LLM Judge. C1 uses a local multimodal MoE judge (*Qwen3.5-35B-A3B-FP8*) [16]. Confidence is the judge's *score*/100 and the unparseable output is assigned the neutral confidence value 0.5. The same pipeline runs on ViDoRe and SÚJB; the prompt, parser, generation config, and per-query raw scores are in the code release. We evaluate a local judge rather than a paid frontier API because much of the retrieved regulatory content is confidential and cannot be sent to third-party services (per-query cost is a secondary concern); our LLM-judge conclusions therefore concern this local-model setting, not the best possible proprietary judge.

Prompt selection. We use a three-shot English prompt, chosen for parse reliability and the best overall AUROC; the tested variants, full prompt text, and per-variant scores are in the code release.

4.4 Hybrid Methods

H1: Tian et al. ECIR'26 (adapted) [19]. Tian's full method augments S3's four score-only features with a cross-encoder relevance score for the top-1 retrieved document. The original uses BERT-QPP [1], a BERT cross-encoder fine-tuned on relevance judgments to regress MRR. We replace it with a *zero-shot* multilingual cross-encoder (`mmarco-mMiniLMv2-L12-H384-v1`) because our SÚJB dataset (Section 3.1) contains Czech queries and no Czech QPP-labeled training data is available, so an English-fine-tuned BERT-QPP would not transfer. H1 should therefore be read as a *conservative* approximation of Tian et al.'s full hybrid, since a fully fine-tuned BERT-QPP would likely strengthen it.

H2: GeneralQPP + LLM Judge (Ours). Our 24 GeneralQPP features augmented with the C1 sufficiency score as a 25th feature. This treats the LLM judgment as an additional signal for the trained predictor rather than a standalone estimator.

H3: Classic Full + LLM Judge. The 18 classic QPP features (S4) augmented with the C1 score as a 19th feature. Table 2 reports the selected model families and per-query latency. Score-only methods are measured on the SÚJB test split; C1 is measured end-to-end over the full SÚJB run at $k = 10$; hybrid latencies add one local LLM-judge call to the corresponding score-based method.

Table 2. Per-query wallclock latency and CV-selected architectures. Arch codes: R = Ridge, L = Logistic Regression, MLP = multilayer perceptron; ViDoRe entries count per-domain picks across the 8 domains (e.g. 8×MLP means an MLP was selected in all 8). C1 end-to-end at $k = 10$, and H2/H3 add one C1 call to the score-based method.

Tier	Method	Ft	Arch (SÚJB)	Arch (ViDoRe)	Latency/query
S0	MaxSim baseline	1	–	–	0.007 ms
S1	GenQPP (ours)	24	R	8×MLP	2.0 ms
S1-Lean	GenQPP-Lean (ours)	13	L	8×MLP	1.9 ms
S2	Huly SIGIR'25	6	L	7×MLP, 1×R	2.8 ms
S3	Tian ECIR'26 (score-only)	4	L	5×MLP, 2×R, 1×L	2.7 ms
S4	Classic Full	18	R	8×MLP	2.7 ms
C1	LLM judge (Qwen3.5-35B)	1	Platt	Platt	6.3 s
H1	Tian ECIR'26 (adapted)	5	L	6×MLP, 1×R, 1×L	982.9 ms
H2	GenQPP + LLM (ours)	25	L	8×MLP	6.3 s
H3	Classic Full + LLM (ours)	19	R	8×MLP	6.3 s

Table 3. Per-domain test AUROC on the eight ViDoRe in-domain settings; the rightmost column reports the weighted average across domains. Best score-only per domain in **bold**, best overall underlined.

	cs	*energy*	*fin en*	*fin fr*	*hr*	*indust*	*pharma*	*physics*	Avg
n_{test}	258	370	371	384	382	340	437	363	2,905
S0	.589	.573	.636	.644	.536	.497	.615	.639	.593
S3	.585	.583	.709	.610	.706	.649	.634	.655	.643
S2	.756	.520	.720	.780	.662	.756	.801	.716	.714
S4	.764	**.835**	.822	.849	.838	.840	**.869**	.837	.835
S1 (ours)	**.905**	.787	**.826**	**.889**	**.848**	.858	.854	**.900**	**.856**
S1-Lean (13 ft, ours)	.809	.748	.797	.776	.824	**.862**	.731	.730	.782
C1	.627	.654	.724	.687	.608	.693	.664	.530	.649
H1	.641	.584	.668	.714	.697	.697	.748	.686	.683
H3	.887	.845	.826	.762	.864	.835	.862	.816	.836
H2 (ours)	.880	.906	.846	.857	.834	.856	.853	.880	.863

5 Results

5.1 Main Results: ViDoRe Benchmark

On the eight ViDoRe domains (14,514 queries total) we evaluate *in-domain*: each domain is split 80/20 and trained and tested within that same domain (vs. the cross-domain LODO setting of Section 6). 3-fold CV *within* the 80% split selects the architecture (11-candidate pool, Section 4.1); the model is then refit on the full 80% and evaluated on the 20% held-out test. Table 3 reports per-domain AUROC and the weighted average, our primary headline result.

Score-based methods. S1 (GeneralQPP) achieves the best weighted-average AUROC of 0.856 (95% CI [0.842, 0.870]) among score-only methods, ahead of S4 (+0.021, S1 wins 6/8 domains), S2 (+0.142), and S3 (+0.213); all three significant under a stratified paired bootstrap (2,000 resamples, cluster-resampled per domain; S1>S4 $p = 0.012$, S1>S2/S3 $p = 0.001$).

Table 4. SÚJB vision-mode results at Hit@5 and Hit@10. Best score-only in **bold**, best overall <u>underlined</u>; C1 values are Platt-calibrated (Section 5.4).

			Hit@5				Hit@10			
Tier	Method	Ft	AUROC	AUPRC	Brier	ECE	AUROC	AUPRC	Brier	ECE
Score	S0 MaxSim	1	.552	.575	.262	.118	.545	.612	.270	.164
	S3 Tian ECIR'26	4	.815	.813	.183	.101	.807	.831	.185	.100
	S2 Huly SIGIR'25	6	.871	.877	.147	.059	.875	.905	.144	.067
	S4 Classic Full	18	.875	.868	.145	.059	.891	.908	.131	.048
	S1 GenQPP (ours)	24	**.893**	.893	.136	.055	**.905**	.923	.124	.061
	S1-Lean (ours)	13	.890	**.894**	**.135**	**<u>.033</u>**	.904	**.923**	**.124**	**<u>.043</u>**
Cont.	C1 LLM Judge	1	.832	.831	.168	.061	.876	.898	.140	.074
Hybrid	H1 Tian (adapted)	5	.820	.820	.179	.087	.809	.839	.184	.102
	H3 Classic + LLM	19	.906	.895	.125	.074	<u>.932</u>	.933	<u>.096</u>	.075
	H2 GenQPP + LLM (ours)	25	<u>.911</u>	<u>.904</u>	<u>.120</u>	.071	.931	<u>.935</u>	.101	.076

Content-based method. C1 (LLM judge) achieves AUROC 0.649 [0.628, 0.670], below all trained score-only methods except S0. The +0.207 gap to S1 is large and statistically significant (paired bootstrap $[+0.184, +0.231]$, $p = 0.001$); for the tested local judge, this weaker discrimination also comes at substantially higher latency (Table 2).

Hybrid methods. H2 (GeneralQPP + LLM judge) reaches AUROC 0.863, only +0.007 over S1 on ViDoRe (n.s., $p = 0.261$) but significant on SÚJB at both depths ($p \leq 0.031$). Crucially, H2 beats H3 (Classic + LLM) by +0.028 ($p = 0.002$), so GeneralQPP adds signal the classic pool lacks even after the LLM score. ViDoRe therefore favors S1 as the latency-efficient choice, with H2 worthwhile only when a local LLM-judge pass is already in the pipeline.

5.2 Diagnostic: Failure-Case Calibration

On ViDoRe, queries with Hit@10=0 range from 21.8% (physics) to 48.4% (HR). C1 assigns mean confidence 0.560 to these queries after the Platt wrapper (Section 4.1) and 0.771 raw, both exceeding MaxSim's 0.467 and confirming systematic LLM overconfidence when the retriever fails. Score-based methods assign substantially lower confidence (S1 and S4 both 0.365); H2 reaches 0.355, the lowest of all methods, showing that the trained predictor downweights C1's overoptimistic scores. Weighted-average positive-minus-negative confidence separation reinforces the pattern: C1 (0.047) vs. S1 (0.394) vs. H2 (0.418). Platt scaling is monotonic; it leaves AUROC unchanged, but it rescales C1's over-confident raw scores onto a calibrated scale, collapsing the apparent positive−negative gap from 0.098 to 0.047 (Fig. 2). For deployment, this matters more than headline AUROC alone: overconfident failure cases are precisely the queries on which abstention is needed.

5.3 Deployment Case Study: SÚJB

Table 4 reports full SÚJB results (302 test queries, 53.0% positive rate at Hit@5) at both retrieval depths. This deployment-oriented case study tests how the methods behave on 500 author-constructed adversarial unanswerable queries under a realistic local-regulatory setup.

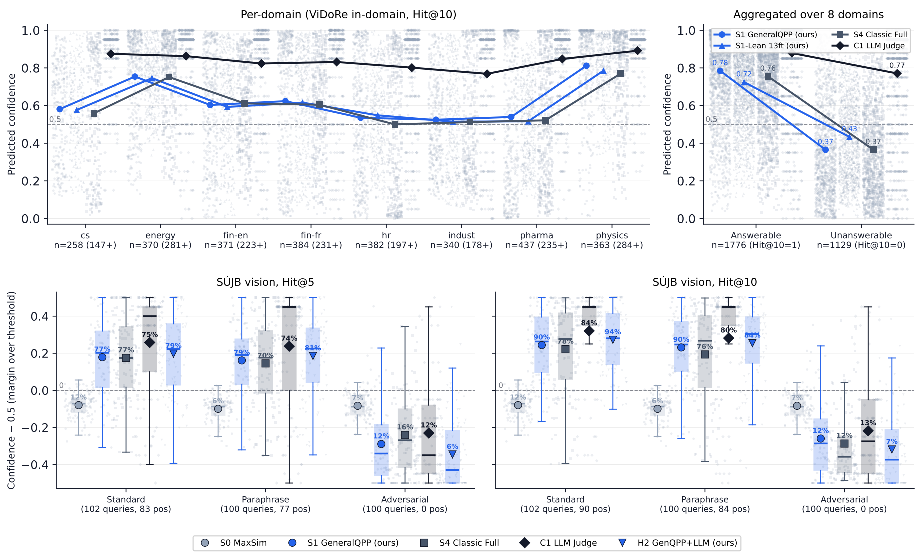

Fig. 2. Per-query predicted confidence. *Top (ViDoRe):* in-domain test splits at Hit@10; left per-domain, right aggregated by ground-truth Hit@10 label. *Bottom (SÚJB):* vision-mode test set by query type at Hit@5 (left) and Hit@10 (right); % annotations give each method's would-answer rate at the 0.5 decision threshold. Across both, score-based predictors place far more mass below the answering threshold on failed/adversarial queries, whereas the C1 LLM judge stays comparatively overconfident.

ViDoRe ranking replicates on SÚJB. The ranking observed on ViDoRe broadly replicates on SÚJB: S1 leads the score-only tier at AUROC 0.893 and improves over S2, S4, and C1 on all four reported metrics. Because the SÚJB test set contains only 302 queries, we treat it primarily as a deployment case study and interpret statistical significance cautiously; the main inferential evidence comes from the much larger ViDoRe benchmark. In sum: on SÚJB, S1 wins score-only; H2 wins on Hit@5 across three of four metrics, while H3 is competitive at Hit@10; S1-Lean provides the best Hit@10 calibration.

Performance by query type. The 302 test queries split into 102 standard answerable, 100 paraphrased (synonym, full-rephrase, and cross-lingual variants), and 100 adversarial unanswerable. This breakdown matters operationally because adversarial unanswerables are the highest-risk queries for a production assistant. We break performance down by query type, separating answerable AUROC (among the 202 answerable queries) from adversarial-detection AUROC (answerable vs. adversarial unanswerable). Adversarial detection is where the methods diverge most: at Hit@5, S1 reaches 0.934 and S1-Lean 0.931, both well above the C1 LLM judge (0.864) and MaxSim (0.489), with the hybrid H2 best overall at 0.954; the ranking holds at Hit@10 (S1 0.949, C1 0.886, H2 0.970). On the answerable queries the methods are closer (AUROC 0.68–0.79). Mean predicted confidence on adversarial queries falls sharply for the score-based and hybrid methods ($\bar{p}_{\mathrm{adv}} \approx 0.15$–$0.21$) but stays high for C1 (≈ 0.27–0.28), the overconfidence this case study targets. S1 achieves this despite seeing only similarity scores (Fig. 2): unanswerable queries produce characteristically flat score distributions that shape features capture reliably, whereas S0 is near-random. Among answerable queries all trained methods cluster closely, so the LLM judge's value lies mostly in in-scope discrimination.

Operating-threshold view. The % annotations in Fig. 2 give each method's would-answer rate at the 0.5 decision threshold. On the 100 adversarial queries, H2 crosses only 6–7% at Hit@5/10 vs. 12% for S1/C1 and 12–16% for S4, roughly halving the best classic baseline's false-answer rate. On answerables, H2's rates (79–94%) track ground-truth positives (77–88%): H2 abstains where the pipeline should reject and answers elsewhere, the asymmetry safety-critical RAG requires.

5.4 Robustness Checks

Method rankings agree across datasets (Spearman $\rho = 0.90$, Kendall $\tau = 0.78$, $p < 0.005$ between the ViDoRe weighted-average and SÚJB Hit@10 AUROC columns), suggesting that the broad method ranking is stable across the benchmark and deployment case study despite differences in language, document type, and annotation.

Platt calibration cuts C1's ECE 6× on ViDoRe (0.249→0.041) and halves it on SÚJB Hit@5 (0.118→0.061) without changing AUROC, so the +0.207 S1−C1 gap reflects discrimination, not calibration.

Train–test AUROC gaps stay below 0.023 for every method reported in Tables 3–4, well below common practical overfitting concern thresholds.

6 Discussion

Score Features vs. Local LLM Judges. For the tested local multimodal judge, the score-distribution methods dominate: S1 outranks C1 on every ViDoRe domain and on SÚJB (Section 5.3), and Section 5.2 shows C1 stays over-confident on retrieval failures while score-distribution methods back off appropriately. S1 thus gives the stronger latency–quality trade-off: H2's AUROC gain over it is small and not significant on ViDoRe, yet still requires a local LLM-judge pass.

LLM as Feature, Not Predictor. Across settings the hybrid that adds the judge as a feature (H2) matches or improves on both the standalone judge (C1) and the score-only predictor (S1), yet its uplift over S1 is small and setting-dependent (Tables 3, 4). The judge therefore pays off mainly when it is already in the inference path: for the tested local multimodal judge, the LLM contributes more as an auxiliary feature than as a standalone predictor.

OOD Robustness: S1-Lean. LODO on ViDoRe drops S1 from 0.856 to 0.706 (0.15 penalty). Forward-backward stepwise peaks at 13 features (LODO AUROC 0.719, *+0.012 over full S1 and +0.032 over classical baseline S4*); S1-Lean trades 0.074 in-domain ViDoRe AUROC for this transfer and for the best SÚJB calibration of any method (ECE 0.043 at Hit@10 vs. S4's 0.048, H2's 0.076). A practical default is to use S1 when modest target-domain labels exist, and S1-Lean when cross-domain transfer matters more.

Limitations. GeneralQPP features were designed on SÚJB training data; ViDoRe mitigates this (S1 wins 6/8 domains) and the S1-Lean LODO result quantifies cross-domain transfer. Validation on live (non-synthetic) queries is the main item of future work (Section 3.1). The SÚJB queries used here were generated rather than collected from production logs, although they were manually spot-checked for relevance. A live-query benchmark for the regulatory client setting is currently being prepared.

With a single retriever and a single local judge in our setup, retriever-agnosticism of the distribution-shape features is an empirical prediction we have not tested directly. C1 is a single local multimodal judge constrained by JSON parsing, and our H1 is a conservative adaptation (zero-shot cross-encoder in place of a fine-tuned BERT-QPP); a stronger fine-tuned hybrid or a frontier cloud judge could narrow or alter the gaps, although such models were outside the local privacy-constrained deployment setting. Accordingly, our conclusions should be read as strongest for local multimodal retrieval-sufficiency prediction with modest target-domain supervision, not as a universal ranking of all RAG confidence estimators.

7 Conclusion

We compared score-based, content-based, and hybrid QPP methods for predicting RAG retrieval sufficiency before generation. The main practical conclusion is that retriever score distributions already provide a strong low-latency signal: on ViDoRe, GeneralQPP is the strongest score-only predictor (0.856 AUROC), ahead of the Classic Full QPP baseline (0.835) and the tested local multimodal LLM judge (0.649), while remaining millisecond-level (Tables 3 and 2). It also remains strong in the private SÚJB case study, where adversarial unanswerable queries are the deployment-relevant failure mode (Table 4).

The tested local multimodal LLM judge is useful, but mainly as an auxiliary feature in selected settings rather than as a standalone confidence estimator. On ViDoRe, adding the judge to GeneralQPP gives only a small and non-significant gain over S1, whereas on SÚJB the hybrid variants are strongest overall. This matters for safety-critical local deployment: score-only predictors can run inline on every query (about 2 ms for GeneralQPP), whereas the tested local LLM judge requires a seconds-level pass ($\sim$6.3 s in our setup; Table 2) and is therefore better reserved for cases where an additional content check is worth the latency.

The results place QPP-style retrieval-sufficiency prediction in the less explored setting of vision-document RAG. The main remaining questions are cross-retriever generalization, validation on live user queries, and stronger local or fine-tuned judge models.

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